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TECHNICAL REPORT

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Abstract

Accurate detection of forest cover change is critical for monitoring ecological health and supporting sustainable land management, particularly in the North Eastern Region of India, where small-scale deforestation and persistent cloud cover pose significant challenges. This study presents a deep learning-based approach for pixel-level forest loss detection using multi-temporal and multi-modal satellite imagery. A Siamese U-Net architecture augmented with Atrous Spatial Pyramid Pooling (ASPP) is employed to capture multi-scale contextual information and effectively model changes between two time periods. The model integrates both Sentinel-1 SAR data, which is robust to cloud interference, and Sentinel-2 optical data, which provides rich spectral information. To address challenges such as class imbalance and sparse change patterns, advanced techniques including attention-based feature refinement, explicit change feature fusion, and deep supervision are incorporated. The proposed approach is evaluated using F1-score and Intersection over Union (IoU), demonstrating improved performance over baseline methods. The results highlight the effectiveness of combining multi-modal data and enhanced architectural design for accurate and reliable forest change detection in complex tropical environments. **Code:**

https://github.com/Harshkumar0403/Group-5-DS-and-AI-Lab-Project

1 Introduction

Tropical forests are among the most ecologically significant biomes on Earth, providing critical ecosystem services including carbon sequestration, watershed protection, biodiversity preservation, and climate regulation. The North Eastern Region (NER) of India — encompassing states such as Meghalaya, Nagaland, Manipur, and Arunachal Pradesh — constitutes one of the world's recognised biodiversity hotspots, harbouring exceptional endemic flora and fauna across its dense tropical and subtropical forests [1]. Despite this ecological importance, these forests face intensifying anthropogenic pressures from infrastructure development, urban expansion, and — most pervasively — traditional shifting cultivation practices known locally as *Jhum* cultivation [2].

Traditional approaches to forest change monitoring rely heavily on optical satellite imagery — most prominently the Landsat-based Hansen Global Forest Change (GFC) dataset, which provides annual global forest loss data at 30-metre spatial resolution [3]. While this product has been transformative for large-scale deforestation monitoring, it suffers from two fundamental limitations in the NER context. First, its 30-metre resolution is insufficient to resolve the small-scale, fragmented Jhum patches that are the dominant mode of deforestation in this region; disturbances spanning only a few pixels at this resolution are systematically underdetected. Second, the NER experiences persistent and heavy cloud cover throughout the monsoon season (June–September), severely limiting the availability of cloud-free optical imagery and introducing substantial temporal data gaps in Landsat-based products [4].

Synthetic Aperture Radar (SAR) imagery offers a compelling solution to the cloud cover limitation. SAR sensors transmit microwave pulses and record backscattered energy, allowing them to penetrate cloud cover and operate independently of solar illumination [5]. The European Space Agency's Sentinel-1 constellation provides freely available, medium-resolution SAR imagery at 10-metre pixel spacing in interferometric wide-swath mode, making it well-suited for operational forest monitoring. However, SAR data alone has limitations in distinguishing vegetation types due to complex backscattering mechanisms. A multi-modal approach combining Sentinel-1 SAR with the high-resolution optical bands of Sentinel-2 thus presents a promising framework for robust, all-weather forest change detection [6].

In this work, we construct a multi-modal dataset fusing Sentinel-1 SAR and Sentinel-2 optical imagery at 10-metre resolution with Hansen-derived binary change labels for Meghalaya and Nagaland, covering 2021–2023. We formulate forest loss detection as a supervised pixel-level binary segmentation task and train a Siamese U-Net augmented with Atrous Spatial Pyramid Pooling [7] as the baseline model. Through controlled experiments we evaluate the impact of Focal Loss [8], data augmentation, pretrained encoders [9], and attention-based decoder enhancements. Our best model achieves a test F1 of 0.5348 and IoU of 0.3650, with a ROC-AUC of 0.9776, on a held-out test set of 867 patches.

2 Objective, Problem Definition and Literature Review

2.1 Objective

The objective of this project is to develop an Artificial Intelligence driven monitoring system for detecting and analyzing forest cover changes in the North-East region of India, specifically focusing on Meghalaya and Nagaland. Satellite imagery provides a reliable and scalable solution for monitoring environmental changes over time.

In this project, multi-source satellite datasets are used to capture both structural and spectral characteristics of vegetation. The dataset is derived from three primary sources: Sentinel-1 Synthetic Aperture Radar (SAR) data, Sentinel-2 multispectral optical imagery, and the Hansen Global Forest Change dataset, which provides labeled information about forest loss.

These datasets are integrated to create a supervised learning framework suitable for training deep learning models for forest change detection.

2.2 Problem Definition

Forest cover change detection is a critical environmental monitoring task that involves identifying spatial and temporal variations in vegetation. Traditional monitoring methods are often manual, time-consuming, and limited in scalability.

The problem addressed in this project is to automatically detect forest loss using bi-temporal satellite imagery. Given two satellite images of the same geographical region captured at different time instances (e.g., 2022 and 2023), the goal is to predict a pixel-wise binary mask indicating areas where forest loss has occurred.

This can be formulated as a supervised semantic segmentation problem:

- Input: Multi-channel satellite image pairs from two time steps
- Output: Pixel-wise binary classification (forest loss / no forest loss)

Key challenges in this problem include:

- Cloud cover and atmospheric disturbances in optical imagery
- Variability in vegetation due to seasonal changes
- Class imbalance (forest loss pixels are sparse)
- Multi-resolution data integration (10m vs 30m)

2.3 Literature Review

Satellite-based forest monitoring has a long history rooted in multispectral imagery. Hansen *et al.* [3] produced the first global, wall-to-wall forest loss product by analysing Landsat time series at 30-metre resolution, establishing the standard reference dataset for deforestation research. However, this product cannot resolve small-scale disturbances or maintain temporal continuity in cloud-prone tropical regions. specifically demonstrated the utility of satellite remote sensing for land cover dynamics in Meghalaya, while Behera *et al.* [2] used Landsat-8 to map Jhum fallows across all eight NER states, finding Manipur and Arunachal Pradesh to be primary hotspots of cyclical disturbance.

The advent of Sentinel-1 SAR addressed the cloud cover limitation. Because SAR sensors transmit microwave energy and record backscattered returns independently of solar illumination, they can observe the Earth's surface through cloud cover and at night [5]. Zhang *et al.* [6] proposed DSNUNet, a double

Siamese branch network specifically designed to fuse Sentinel-1 SAR and Sentinel-2 optical imagery for forest change detection, showing that the two modalities are strongly complementary. Xiang *et al.* [10] further demonstrated that deep learning applied to Sentinel-2 time series achieves accurate dynamic forest change mapping at the provincial scale.

The U-Net architecture [11], originally designed for biomedical image segmentation, became a foundational backbone for pixel-level remote sensing prediction owing to its encoder-decoder design with skip connections that preserve fine-grained spatial detail. Daudt *et al.* [12] extended this idea to the change detection setting by proposing Fully Convolutional Siamese Networks (FC-Siam), in which weight-shared encoder branches process each temporal image independently, ensuring that features from both time steps lie in a consistent representation space. Chen *et al.* [13] further combined the Siamese paradigm with U-Net and attention mechanisms, demonstrating improved performance on high-resolution change detection benchmarks.

Deep residual networks [9] have become the dominant choice of pretrained encoder backbone in remote sensing transfer learning due to their strong ImageNet representations of edges, textures, and mid-level patterns. However, the domain gap between natural RGB imagery and multi-modal SAR+multispectral satellite data remains a significant challenge when adapting pretrained weights to non-standard input channels [14].

3 Dataset Preparation

3.1 Overview

The dataset used in this project is not directly available as a ready-to-use benchmark dataset. Therefore, a custom dataset was created by integrating multi-source satellite imagery and forest loss labels. The dataset combines information from Sentinel-1 radar data, Sentinel-2 optical imagery, and the Hansen Global Forest Change dataset.

The goal of this dataset preparation process is to generate aligned input-label pairs suitable for supervised deep learning models for forest change detection.

3.2 Data Sources

3.3 Study Area

The study focuses on two states in the North Eastern Region (NER) of India: **Meghalaya** and **Nagaland**. Both regions are characterised by dense tropical and subtropical broadleaf forests, complex hilly terrain, and a high prevalence of Jhum (shifting) cultivation — a slash-and-burn agricultural practice that is the primary driver of small-scale, fragmented deforestation in the NER [2]. These characteristics make this region a particularly demanding testbed for forest change detection models, and a region underserved by existing global products at 30-metre resolution [3].

3.4 Data Sources

3.4.1 Sentinel-1 SAR Imagery

Sentinel-1 (ESA) provides C-band SAR data, enabling cloud-penetrating and day–night Earth observation [5]. Level-1 GRD products in Interferometric Wide (IW) mode (250 km swath, $\sim$10 m resolution) were used [15]. Two dual-polarisation channels were selected: VV (surface and structural response) and VH (vegetation volume scattering).

Preprocessing was performed in Google Earth Engine (GEE), including thermal noise removal, radiometric calibration (sigma-nought), and terrain correction using SRTM DEM, with outputs in dB scale [16].

3.4.2 Sentinel-2 Optical Imagery

Sentinel-2 (ESA) provides multispectral imagery across 13 bands. Level-2A surface reflectance data (atmospherically corrected via Sen2Cor) were used [17]. Four 10 m bands were selected: B2 (Blue), B3 (Green), B4 (Red), and B8 (NIR), capturing key vegetation characteristics [18].

To reduce cloud effects and seasonal noise, 3-month median composites were generated.

3.4.3 Hansen Global Forest Change Labels

Ground truth labels were derived from the Hansen Global Forest Change dataset (30 m resolution) [3]. Forest loss (2021–2023) was binarised into change (1) and no-change (0). Resampling to 10 m introduces boundary uncertainty, leading to label noise, which limits achievable model accuracy [4].

3.5 Data Extraction and Processing

Satellite data was accessed and processed using Google Earth Engine. The region of interest was defined over the North-East region of India, specifically Meghalaya and Nagaland.

Bi-temporal imagery corresponding to two time periods (2022 and 2023) was extracted. Relevant spectral and radar bands were selected and preprocessed.

Since Sentinel-2 imagery is available at 10-meter resolution and the Hansen dataset is at 30-meter resolution, the Hansen dataset was resampled to 10-meter resolution using nearest neighbor interpolation to ensure pixel-wise alignment between inputs and labels.

3.6 Patch Generation

The extracted satellite images were divided into smaller patches to make them suitable for deep learning models.

Each sample in the dataset consists of:

- Input tensor of size $6 \times 256 \times 256$
- Corresponding label mask of size $1 \times 256 \times 256$

The six channels correspond to:

- Sentinel-1 VV
- Sentinel-1 VH
- Sentinel-2 B2
- Sentinel-2 B3
- Sentinel-2 B4
- Sentinel-2 B8

This patch-based approach increases the number of training samples and allows efficient model training.

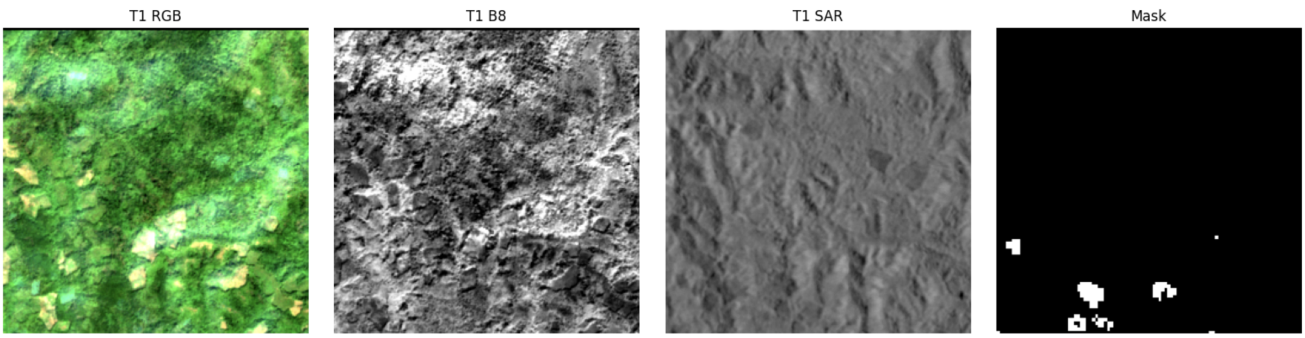


Figure 1: Band Visualization

3.7 Dataset Size

Using the above pipeline, a total of **7500 training samples** were generated. Each sample represents a unique spatial region within the study area.

The dataset is structured as paired input-output samples, where each input corresponds to a multi-channel satellite image and the output represents the forest loss mask.

3.8 Data Quality and Sanity Checks

Before finalizing the dataset, a series of sanity checks were performed to ensure data quality. These include:

- Removal of patches with excessive cloud coverage
- Verification of alignment between satellite imagery and label masks
- Inspection of abnormal or missing pixel values
- Filtering out patches dominated by non-forest regions such as water bodies or urban areas

These preprocessing steps ensure that the dataset is clean, consistent, and suitable for training deep learning models.

3.9 Patch Extraction Strategy

To facilitate efficient training and increase the number of training samples, large satellite images are divided into smaller patches of size 256×256. This patch-based approach allows the model to focus on localized patterns of change while maintaining manageable computational requirements. It also increases dataset diversity and improves generalization.

3.10 Data Normalization

All input channels are normalized using dataset-specific mean and standard deviation values. This standardization ensures that each channel has a comparable scale, which stabilizes training and accelerates convergence. Normalization is particularly important in multi-modal settings where different data sources (SAR and optical) have varying value ranges.

3.11 Class Imbalance Analysis

Forest change detection datasets are inherently imbalanced, with the majority of pixels belonging to the no-change class. In the constructed dataset, change pixels represent a small fraction of the total pixel distribution. This imbalance can bias the model toward predicting the majority class, leading to poor detection performance on minority (change) regions. Addressing this imbalance is a key consideration in model design and loss function selection.

3.12 Change Ratio Distribution and Binning

To better understand the distribution of change across samples, the ratio of change pixels to total pixels is computed for each patch. Based on this ratio, samples are categorized into four bins: no change, very sparse

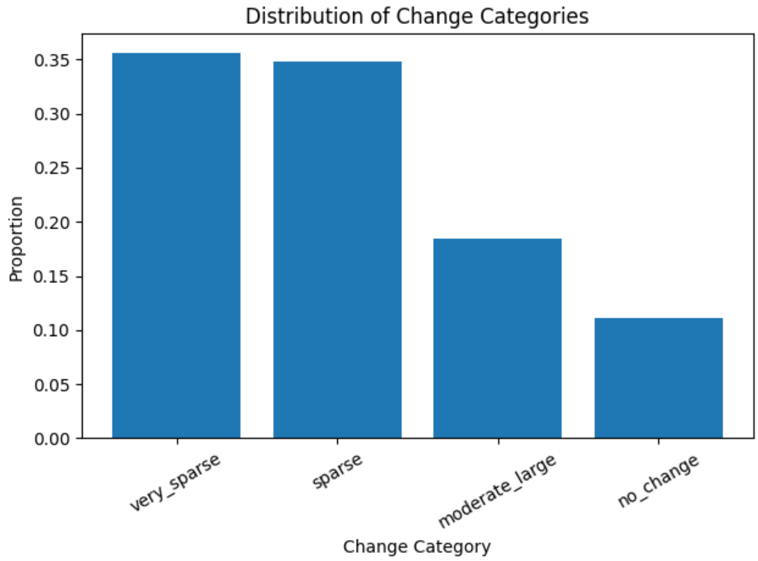


Figure 2: Distribution of samples across change intensity bins showing class imbalance in the dataset.

change, sparse change, and moderate-to-large change. This binning strategy enables stratified sampling and ensures that the dataset captures a balanced representation of different change intensities.

3.13 Train-Validation-Test Split Strategy

The dataset is split into training, validation, and test sets using a stratified sampling approach based on region and change intensity bins. This ensures that each split maintains a similar distribution of change patterns, preventing bias toward specific regions or change types. A typical split ratio of 70% training, 15% validation, and 15% testing is used to balance model training and evaluation.

3.14 Final Dataset Structure

The final dataset consists of structured input-label pairs ready for model training. Each sample captures both spectral and structural characteristics of vegetation along with corresponding forest loss labels.

This custom dataset forms the foundation for training and evaluating the proposed deep learning model for forest change detection.

4 Model Architecture

4.1 Motivation and Model Choice

Forest cover change detection is formulated as a **bi-temporal semantic segmentation problem**, where the objective is to detect pixel-wise changes between two satellite images captured at different time instances.

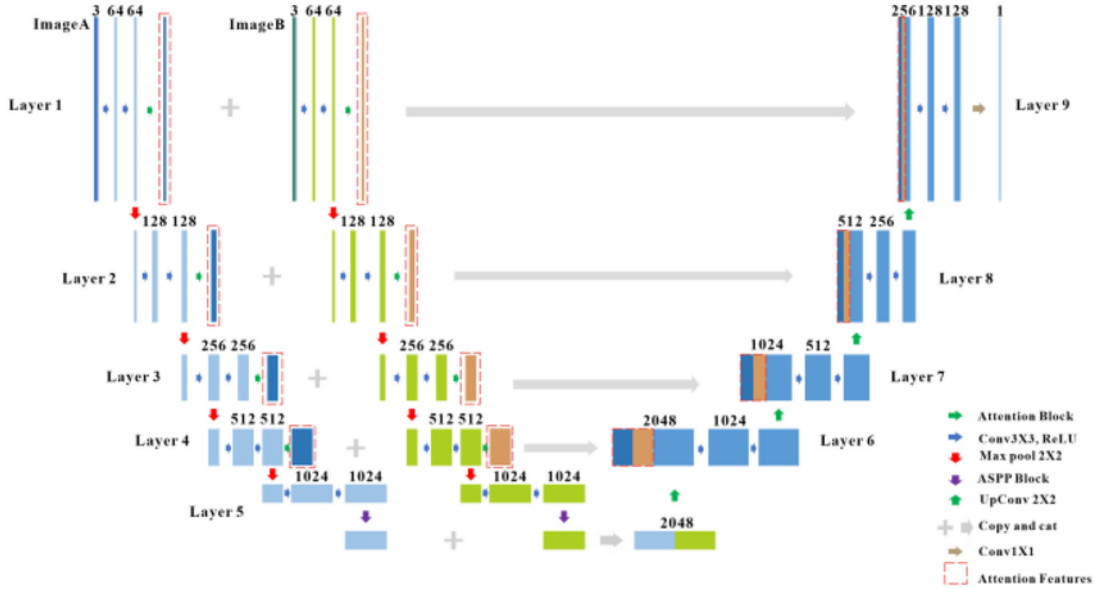


Figure 3: Structure of baseline model

Stage	Layer	Input Channels	Output Channels
Encoder			
Conv Block 1	DoubleConv	6	64
Conv Block 2	DoubleConv	64	128
Conv Block 3	DoubleConv	128	256
Conv Block 4	DoubleConv	256	512
Pooling	MaxPool	–	–
ASPP Module			
1×1 Conv	Conv2d	512	512
3×3 Conv (d=6)	Conv2d	512	512
3×3 Conv (d=12)	Conv2d	512	512
3×3 Conv (d=18)	Conv2d	512	512
Projection	Conv2d	2048	512
Decoder			
UpBlock 4	TransposeConv + DoubleConv	1536	512
UpBlock 3	TransposeConv + DoubleConv	768	256
UpBlock 2	TransposeConv + DoubleConv	384	128
UpBlock 1	TransposeConv + DoubleConv	192	64
Output			
Final Layer	Conv2d (1×1)	64	1

Figure 4: Architecture of Siamese U-Net with ASPP

To address this, a **Siamese U-Net architecture integrated with an Atrous Spatial Pyramid Pooling (ASPP) module** is used. This architecture is specifically designed to:

- Learn spatial features from multi-channel satellite imagery
- Capture temporal differences between two time instances
- Perform accurate pixel-wise segmentation

A key aspect of the model is the use of **feature differencing** between encoded representations of the two temporal inputs, which allows the network to explicitly focus on changes rather than absolute features.

4.2 Reference Architectures

The proposed model is inspired by the following foundational works:

- U-Net Architecture: U-Net: Convolutional Networks for Biomedical Image Segmentation
- DeepLab (ASPP Module): DeepLab: Semantic Image Segmentation with Atrous Convolution
- Siamese Networks for Change Detection: Fully Convolutional Siamese Networks for Change Detection

5 Methodology

5.1 Problem Formulation

Forest cover change detection is formalised as a supervised binary semantic segmentation task. Given a pair of co-registered multi-modal images $T_1, T_2 \in R^{C\times H\times W}$ acquired at times t_1 (2021) and t_2 (2023), the goal is to learn a mapping:

$$f_\theta : (T_1, T_2) \mapsto \hat{Y} \in [0,1]^{H\times W} \tag{1}$$

where $\hat{Y}_{i,j}$ is the predicted probability of forest loss at pixel (i, j), and θ denotes the learnable parameters. The final binary prediction is obtained by thresholding: $\hat{Y}^{(b)}_{i,j} = \mathbf{1}[\hat{Y}_{i,j} \geq \tau]$, where τ is the decision threshold optimised on the validation set.

5.2 Baseline Model: Siamese U-Net with ASPP

5.2.1 U-Net Backbone

The baseline architecture is built upon the U-Net [11], an encoder-decoder network with skip connections that bridges corresponding spatial levels of the encoder and decoder. Each skip connection concatenates encoder feature maps with the corresponding upsampled decoder feature maps, allowing the decoder to recover fine-grained spatial detail lost during progressive downsampling. The use of skip connections is particularly important in change detection, where sharp deforestation boundaries and small-scale patches must be precisely localised.

5.2.2 *Siamese Encoder with Shared Weights*

To process the bitemporal input, the Siamese paradigm is adopted [12]. Two encoder branches — one for T_1 and one for T_2 — share identical weights throughout training. This weight-sharing enforces a *consistent representation space*: features extracted from both time steps are governed by the same learned filters, ensuring that temporal differences in the feature space are attributable to genuine scene changes rather than to encoder divergence. The encoder consists of four `DoubleConv` blocks, each comprising two consecutive Conv2d – BatchNorm – ReLU operations [19], followed by 2×2 max pooling for spatial downsampling. The channel progression is: $6 \to 64 \to 128 \to 256 \to 512$, producing multi-scale feature maps at four spatial resolutions. Formally, for each encoder stage l:

$$E_l^{(1)} = \text{DoubleConv}_l(T_1), \quad E_l^{(2)} = \text{DoubleConv}_l(T_2) \tag{2}$$

where DoubleConv_l uses the same weights for both branches.

5.2.3 *Atrous Spatial Pyramid Pooling (ASPP)*

A critical design choice for detecting forest loss events of widely varying spatial extent is the incorporation of an ASPP module [7] at the encoder bottleneck (deepest layer). Jhum clearances in the NER range from large contiguous openings spanning thousands of pixels to narrow linear patches only a few pixels wide. A standard single-scale bottleneck convolution cannot simultaneously capture context at both scales within a single receptive field. ASPP resolves this by applying four parallel atrous (dilated) convolutions with dilation rates $r \in \{1, 6, 12, 18\}$ to the 512-channel bottleneck feature map:

$$\mathbf{F}_{\text{ASPP}} = \text{Conv}_{1\times 1}\left(\left[\mathbf{f}^{(r=1)} \,\|\, \mathbf{f}^{(r=6)} \,\|\, \mathbf{f}^{(r=12)} \,\|\, \mathbf{f}^{(r=18)}\right]\right) \tag{3}$$

where $\|$ denotes channel-wise concatenation. Each $\mathbf{f}^{(r)}$ is the output of a 3×3 atrous convolution with dilation rate r applied to the bottleneck features, capturing context within receptive fields of increasing size without sacrificing spatial resolution. The concatenated 2048-channel tensor is projected back to 512 channels via a 1×1 convolution, maintaining a compact representation for the decoder.

5.2.4 *Decoder and Skip Connection Fusion*

The decoder consists of four `UpBlock` modules arranged symmetrically to the encoder. Each `UpBlock` applies a 2×2 transposed convolution for $2\times$ upsampling, followed by a `DoubleConv` block. At each decoder level, the upsampled feature map is concatenated with *both* the T1 and T2 encoder skip features from the corresponding encoder stage:

$$D_l = \text{DoubleConv}_l\left(\text{Up}(D_{l+1}) \parallel E_l^{(1)} \parallel E_l^{(2)}\right) \tag{4}$$

This joint skip connection provides the decoder with direct access to unprocessed temporal features from both epochs, enabling it to compare fine-grained spatial detail at multiple scales rather than relying solely on the compressed bottleneck representation. The decoder channel progression is: $512 \to 256 \to 128 \to 64$. A final 1×1 convolution maps the 64-channel decoder output to a single-channel logit map, which is passed through a sigmoid activation to produce per-pixel change probabilities.

The complete architecture is summarised in Table **??**.

Stage	**Layer**	**Input Ch.**	**Output Ch.**
Encoder	DoubleConv Block 1	6	64
	DoubleConv Block 2	64	128
	DoubleConv Block 3	128	256
	DoubleConv Block 4	256	512
ASPP	$4\times$ Atrous Conv (r=1,6,12,18)	512	4×512
	Projection Conv 1×1	2048	512
Decoder	UpBlock 4 (TransConv + DoubleConv)	1536	512
	UpBlock 3 (TransConv + DoubleConv)	768	256
	UpBlock 2 (TransConv + DoubleConv)	384	128
	UpBlock 1 (TransConv + DoubleConv)	192	64
Output	Conv 1×1 + Sigmoid	64	1

Table 1: Siamese U-Net with ASPP architecture summary.

5.3 Model Enhancement Variants

Beyond the baseline, three enhancement variants were investigated in controlled experiments.

5.3.1 Training-Level Improvements

The first variant retained the baseline architecture while improving the training pipeline:

1. **Loss function refinement**: The BCE+Dice combination was replaced by a weighted Dice+Focal formulation ($0.7 \cdot \mathscr{L}_{\text{Dice}} + 0.3 \cdot \mathscr{L}_{\text{Focal}}$), as detailed in Section 5.4.
2. **Data augmentation**: Random flips and 90° rotations were applied during training (Section 2.4).
3. **Dynamic threshold optimisation**: Rather than a fixed threshold of 0.5, the optimal decision threshold τ^* was selected by searching over $[0.05, 0.55]$ in steps of 0.05 to maximise F1-Score on the validation set at the end of training.
4. **Learning rate scheduling**: A `ReduceLROnPlateau` scheduler monitored validation IoU with a re-

duction factor of 0.5 and patience of 2 epochs.

5.3.2 *Pretrained ResNet-34 Encoder*

The second variant replaced the from-scratch VGG-style encoder with a pretrained ResNet-34 backbone [9] while keeping the ASPP bottleneck and all decoder blocks unchanged. ResNet-34 was selected because its stage output channel dimensions (64, 64, 128, 256, 512) match the original encoder progression exactly, requiring no architectural changes downstream. The encoder was organised into five explicit stages: `enc0` (conv1+BN+ReLU, 64ch), `enc1` (maxpool+layer1, 64ch), `enc2` (layer2, 128ch), `enc3` (layer3, 256ch), and `enc4` (layer4, 512ch).

The primary challenge was adapting the ImageNet-pretrained first convolution (3 input channels) to the 6-channel satellite input. This was resolved by initialising the new Conv2d$(6, 64, 7\times 7)$ weights by averaging the pretrained 3-channel kernel across the input dimension and tiling the result across all 6 channels, thereby preserving the pretrained spatial filter structure in deeper layers while adapting only the input layer [14].

5.3.3 *SiameseUNet V2 with Attention, Change Fusion, and Deep Supervision*

The third variant introduced three structural improvements into the decoder path while leaving the encoder and ASPP module unchanged:

Temporal Bottleneck Fusion. Before entering the ASPP, features from both temporal branches are concatenated to 1024 channels and projected back to 512 channels via a 1×1 Conv-BN-ReLU block, ensuring the ASPP receives a jointly conditioned temporal representation rather than processing each time step independently.

Attention Gates and Change Feature Fusion. Each decoder skip connection is processed by a dedicated Attention Gate [20] that uses the upsampled decoder feature as a gating signal to produce a soft spatial mask suppressing background activations. Since approximately 89% of pixels are unchanged, this suppression is essential for preventing the decoder from wasting representational capacity on uninformative background regions. The attended skip features are then passed through a `ChangeFusion` module that explicitly computes four interaction terms:

$$\mathbf{c} = [t_1 \parallel t_2 \parallel |t_1 - t_2| \parallel t_1 \odot t_2] \tag{5}$$

where t_1, t_2 are the attended T1 and T2 skip features, $\odot$ is element-wise multiplication, and $\parallel$ denotes channel concatenation to 4×ch. A 1×1 projection then reduces back to 2×ch, providing the decoder with explicit change magnitude ($|t_1 - t_2|$) and feature correlation ($t_1 \odot t_2$) signals.

Deep Supervision. Three auxiliary 1×1 convolutional heads are attached to intermediate decoder outputs D_4, D_3, and D_2, each producing a full-resolution prediction [21]. The total training loss is:

$$\mathcal{L}_{\text{total}} = 1.0 \cdot \mathcal{L}_{\text{main}} + 0.4 \cdot \mathcal{L}_{\text{aux4}} + 0.2 \cdot \mathcal{L}_{\text{aux3}} + 0.1 \cdot \mathcal{L}_{\text{aux2}} \tag{6}$$

Deep supervision injects gradient signal directly into early decoder layers, addressing the training plateau observed between epochs 25–30 in preliminary runs. All auxiliary heads are deactivated at inference time via a `self.training` guard, adding zero computational overhead during evaluation.

5.4 Loss Functions

5.4.1 Dice Loss

Dice Loss optimises the pixel-level overlap between the predicted probability map $\hat{Y}$ and the binary ground truth mask Y, making it naturally robust to class imbalance by focusing on the positive class overlap [22]:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i \hat{Y}_i Y_i}{\sum_i \hat{Y}_i + \sum_i Y_i + \varepsilon} \tag{7}$$

where $\varepsilon = 10^{-6}$ is a smoothing term to prevent division by zero. By optimising the ratio of intersection to union, Dice Loss penalises missed detections (false negatives) and false alarms (false positives) symmetrically, which is desirable for the sparse forest loss signal.

5.4.2 Focal Loss

Focal Loss [8] extends standard Binary Cross-Entropy by introducing a modulating factor $(1 - p_t)^\gamma$ that down-weights the contribution of well-classified easy examples and concentrates gradient updates on hard, ambiguous samples:

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p_t)^\gamma \log(p_t) \tag{8}$$

where $p_t = \hat{Y}_i$ if $Y_i = 1$ and $p_t = 1 - \hat{Y}_i$ otherwise. The parameters $\alpha = 0.8$ and $\gamma = 2$ were used throughout all experiments. The class weighting factor α further addresses imbalance by up-weighting the minority forest loss class, while $\gamma = 2$ ensures that confidently predicted background pixels contribute negligibly to the gradient.

5.4.3 Combined Dice + Focal Loss

Based on preliminary experiments, the composite loss:

$$\mathcal{L} = 0.7 \cdot \mathcal{L}_{\text{Dice}} + 0.3 \cdot \mathcal{L}_{\text{Focal}} \tag{9}$$

was adopted for all main experiments. The Dice component dominates (weight 0.7) to preserve recall on sparse change pixels, while the Focal component (weight 0.3) suppresses background noise and stabilises early training dynamics. This formulation consistently outperformed both BCE+Dice and Focal-only baselines in preliminary ablation runs.

5.5 Evaluation Metrics

All metrics are computed at the pixel level on binary predictions obtained by thresholding sigmoid-activated logits at the optimal threshold τ^*. Let TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives respectively.

Precision and Recall.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{10}$$

F1-Score. The harmonic mean of precision and recall, used as the *primary* evaluation metric due to its insensitivity to the large pool of true negatives in imbalanced data:

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

Intersection over Union (IoU). Used as the secondary metric and the model checkpointing criterion:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{12}$$

IoU is a stricter measure than F1, as it penalises both false positives and false negatives without the compensating effect of true negatives.

ROC-AUC and PR-AUC. Area under the Receiver Operating Characteristic curve and Precision-Recall curve respectively, computed from the continuous sigmoid outputs without thresholding. PR-AUC is more informative than ROC-AUC under extreme class imbalance, as the PR curve does not benefit from the large pool of true negatives that inflates the ROC curve.

Decision Threshold Selection. Rather than using a fixed threshold of 0.5, the optimal τ^* is searched over $[0.05, 0.55]$ in steps of 0.05 on the validation set at the end of training, selecting the value that maximises F1-Score. Class imbalance typically shifts the optimal threshold below 0.5, as the model assigns systematically lower confidence scores to the minority class.

5.6 Training Setup

5.6.1 Hardware and Framework

All models were trained on a Kaggle computational notebook using an NVIDIA Tesla T4 GPU (15 GB VRAM). Experiments were implemented in PyTorch with Automatic Mixed Precision (AMP) training via `torch.amp.autocast` and `GradScaler` [23]. AMP performs forward passes in float16 while maintaining float32 precision for gradient accumulation and parameter updates, reducing memory consumption and accelerating matrix operations on Tensor Core hardware.

5.6.2 Optimiser and Scheduler

The Adam optimiser [24] was used with an initial learning rate of 1×10^{-4} and momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$. A `ReduceLROnPlateau` scheduler monitored validation IoU with a reduction factor of 0.5 and patience of 2 epochs, halving the learning rate whenever no validation improvement was observed. Early stopping with patience 5 prevented overfitting and reduced unnecessary compute. All experiments used a batch size of 16 and were allowed a maximum of 50 epochs, with the best checkpoint selected by highest validation IoU.

5.6.3 Hyperparameter Summary

The fixed hyperparameter configuration used across all experiments is summarised in Table 2.

Table 2: Fixed hyperparameter configuration across all experiments.

Hyperparameter	**Value**
Optimiser	Adam
Initial learning rate	1×10^{-4}
(β_1, β_2)	$(0.9, 0.999)$
Batch size	16
Maximum epochs	50
LR scheduler	ReduceLROnPlateau
LR reduction factor	0.5
Scheduler patience	2 epochs
Early stopping patience	5 epochs
Mixed precision	AMP (float16 / float32)
Input patch size	256×256 pixels
Input channels	6 (S1-VV, S1-VH, S2-B2, B3, B4, B8)
Loss function	$0.7 \cdot \mathscr{L}_{\text{Dice}} + 0.3 \cdot \mathscr{L}_{\text{Focal}}$
Decision threshold	Dynamic search over $[0.05, 0.55]$

5.6.4 Experiment Tracking

All experiments were tracked using Weights & Biases (W&B) [25], logging training loss, validation F1, validation IoU, and learning rate at each epoch. Best model checkpoints were saved as W&B artefacts to ensure reproducibility. Each experiment was run independently with identical hyperparameters and identical data splits, modifying only the component under study (architecture or loss function), ensuring that all observed performance differences are attributable solely to the modification being evaluated.

6 Results and Analysis

This section presents a comprehensive evaluation of all experiments conducted in this study. Results are reported across four dimensions: (i) baseline model performance and training dynamics; (ii) incremental gains from each enhancement variant; (iii) held-out test set evaluation with full metric breakdown; and (iv) per-category analysis, qualitative visualisations, and systematic error characterisation.

6.1 Baseline Model Performance

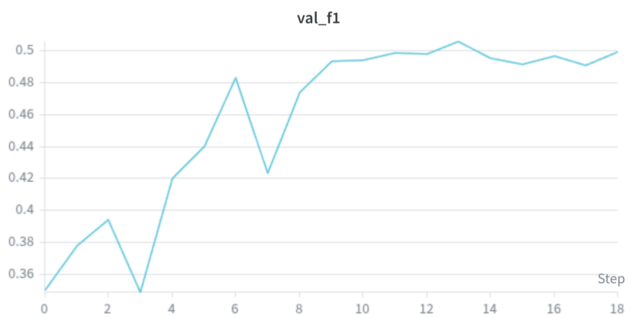
The baseline Siamese U-Net with ASPP, trained from scratch using a combined BCE and Dice loss ($0.5 \cdot \mathscr{L}_{\text{Dice}} + 0.5 \cdot \mathscr{L}_{\text{BCE}}$), served as the quantitative reference point against which all subsequent enhancements were evaluated. Training was run for a maximum of 50 epochs with early stopping (patience = 5), and the best checkpoint was selected according to peak validation IoU.

6.1.1 Training Summary

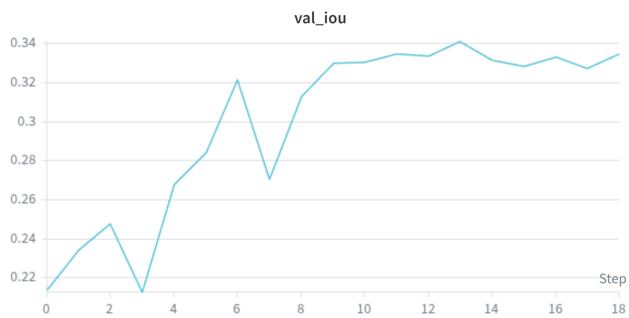
Table 3 reports the key training statistics for the baseline model. Training terminated at epoch 19, with the best checkpoint recorded at epoch 14, indicating that validation performance saturated well before the maximum epoch budget.

Table 3: Baseline model training summary.

Metric	Value
Best Validation F1	0.5059
Best Validation IoU	0.3413
Final Training Loss	0.3517
Epoch of Best Checkpoint	14
Total Epochs Trained	19
Early Stopping Triggered	Yes (patience = 5)



(a) Validation F1 Score



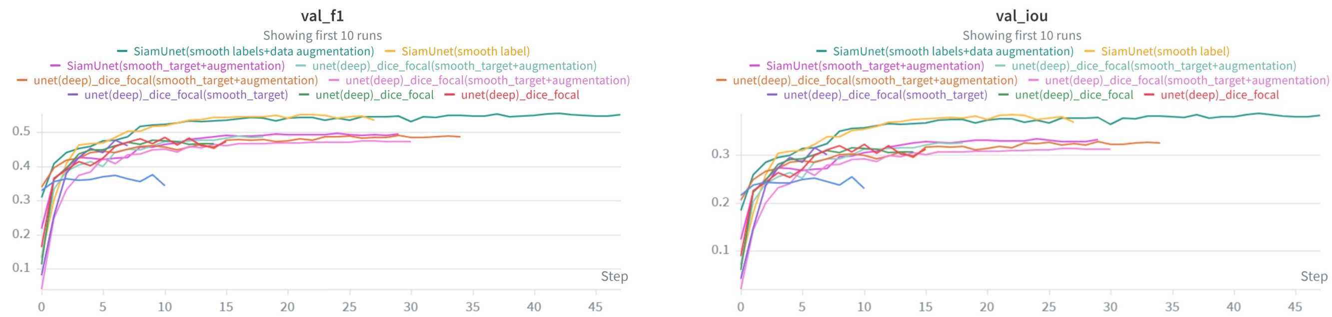
(b) Validation IoU

6.1.2 Training Dynamics

Analysis of the validation curves reveals several characteristic patterns. Both validation F1 and IoU exhibit a broadly increasing trend across the first 14 epochs, punctuated by notable instability in the early training phase. A sharp drop is observed at epoch 3, followed by recovery, and again at epoch 7 where F1 fell from 0.483 to 0.424 and IoU fell from 0.321 to 0.271 before recovering in the following epoch.

This oscillatory behaviour in early training is characteristic of from-scratch initialisation on a small, class-imbalanced dataset. The model alternates between over-predicting the majority background class (resulting in low recall and low F1) and recovering towards the minority forest loss class (resulting in higher recall but elevated false positives). From epoch 9 onward, both curves stabilise and plateau in a narrow band — F1 between 0.491 and 0.506, IoU between 0.328 and 0.341 — indicating that the model has reached the limit of its representational capacity under the current configuration.

The training loss decreases monotonically from 0.753 to 0.352, confirming that the model continues to fit the training data after validation performance saturates. This train-validation divergence is consistent with the early stopping trigger and motivates the architectural and training-level improvements explored in subsequent experiments.



(a) Validation F1 Score on various experiments (b) Validation IoU score on various experiments

6.2 Enhancement Experiment: Data Augmentation and Loss Refinement

Building directly on the baseline, the first enhancement experiment introduced random horizontal and vertical flips, 90° rotations, a refined Dice+Focal loss ($0.7 \cdot \mathscr{L}_{\text{Dice}} + 0.3 \cdot \mathscr{L}_{\text{Focal}}$), dynamic threshold optimisation, and a `ReduceLROnPlateau` scheduler. All other hyperparameters were held constant. Table 4 presents the performance comparison.

Table 4: Performance comparison: Baseline vs. Enhanced model.

Model	Best Val F1	Best Val IoU	Epoch to Best	Total Epochs
Baseline (BCE+Dice)	0.5060	0.3413	14	19
Enhanced (Dice+Focal + Aug.)	0.5189	0.3526	23	30

6.2.1 Analysis

The enhanced model consistently outperforms the baseline across all epochs in both F1 and IoU. The most striking difference is in the early training phase: the enhanced model reaches a validation IoU of approximately 0.27 by epoch 1 and F1 of 0.42 by epoch 3, whereas the baseline required several additional epochs

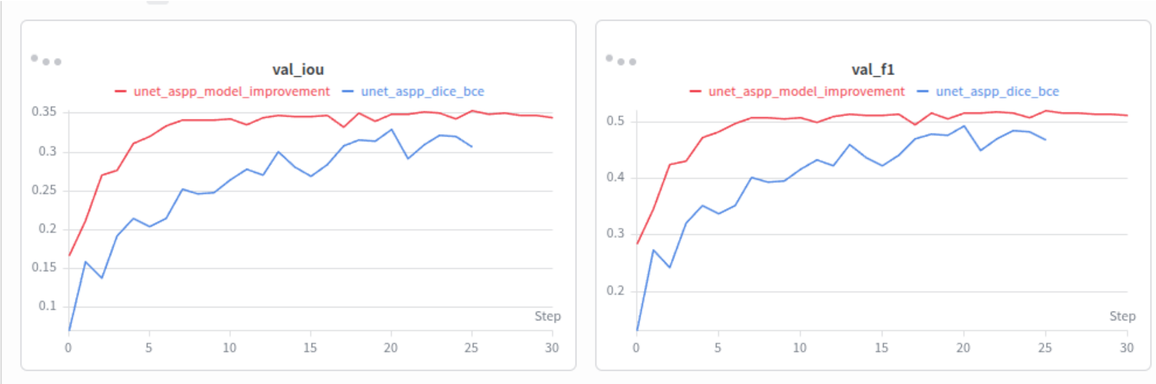


Figure 5: Validation IoU (left) and F1 (right) comparison between the baseline `unet_aspp_dice_bce` (blue) and the enhanced model `unet_aspp_model_improvement` (red).

to achieve comparable values. This faster initial convergence is attributable to the Focal Loss component directing gradient updates towards hard positive forest loss pixels from the first iteration, combined with augmentation introducing more varied spatial configurations per epoch.

However, both models converge to a similar performance ceiling. The enhanced model improves F1 by only 0.013 and IoU by 0.011 over the baseline. The plateau now occurs around epoch 23–25 rather than epoch 14, suggesting that the training-level improvements delay saturation but do not fundamentally overcome the representational bottleneck imposed by the from-scratch encoder. This observation directly motivates the architectural experiments described in the following subsections.

6.3 Enhancement Experiment: Pretrained ResNet-34 Encoder

The second experiment replaced the from-scratch VGG-style encoder with a pretrained ResNet-34 backbone [9] while keeping the ASPP bottleneck and all four decoder UpBlocks unchanged. The results, reported in Table 5, reveal a counterintuitive outcome: despite the stronger ImageNet initialisation, the pretrained encoder model *underperformed* relative to the baseline.

The model achieved a best validation F1 of 0.4625 and IoU of 0.3026 at epoch 12, after which early stopping was triggered at epoch 17. The training loss decreased steadily from 0.7115 to 0.2706, yet validation metrics plateaued and began oscillating after epoch 12, indicating a growing train-validation gap. Three factors likely explain this outcome:

1. **Domain gap**: ImageNet-pretrained features are conditioned on natural RGB imagery. Our 6-channel input (Sentinel-1 SAR + Sentinel-2 optical) exhibits fundamentally different statistical properties — SAR backscatter follows a multiplicative speckle noise model and optical reflectance is spectrally non-overlapping with RGB [14].
2. **Weight initialisation noise**: Adapting the 3-channel pretrained first convolution to 6 channels by averaging and tiling the pretrained kernel may have introduced structural inconsistencies that propagate

into deeper layers.

3. **Insufficient training data**: With only 5,250 training patches, the model lacks sufficient signal to fine-tune the large number of pretrained ResNet-34 parameters effectively within 17 epochs.

6.4 Enhancement Experiment: SiameseUNet V2 with Attention, Change Fusion, and Deep Supervision

The third experiment introduced temporal bottleneck fusion, attention-gated skip connections, explicit change feature fusion (`ChangeFusion`), and deep supervision into the decoder path, while leaving the encoder and ASPP unchanged. Despite the substantial architectural additions, the model achieved a best validation F1 of 0.4869 and IoU of 0.3240 at epoch 12, with early stopping at epoch 17.

The higher initial training loss (1.08 at epoch 1) reflects the additional complexity from the deep supervision auxiliary losses, and epoch times were approximately 3$\times$ longer ($\approx$3m 40s vs. $\approx$1m for the baseline) due to the attention and fusion modules. While the model exhibited a steady improvement trajectory up to epoch 12, it ultimately failed to surpass the enhanced baseline. This outcome suggests that, at the 7,500-sample scale of the constructed dataset, the added modules increase optimisation difficulty without sufficient training data to offset the higher parameter count.

6.5 Overall Model Comparison

Table summarises the performance of all four experimental configurations under consistent evaluation conditions.

Model	Best F1	Best IoU	Best Epoch	Total Epochs
Baseline (BCE+Dice)	0.5060	0.3413	14	19
Enhanced (Dice+Focal + Aug.)	**0.5189**	**0.3526**	23	30
Pretrained ResNet-34	0.4625	0.3026	12	17
SiameseUNet V2 (Attn+CF+DS)	0.4869	0.3240	12	17

Table 5: Overall comparison of all experimental configurations on the validation set. Best model highlighted in gold; models below baseline highlighted.

The key finding is that the two most complex architectural variants both *underperform* the simple baseline, while the lightweight training-level enhancements provide the most consistent gains. Collectively, this pattern suggests that for this dataset scale, **training stability and loss design have a greater impact on performance than architectural complexity**. Future improvements should prioritise dataset expansion and semi-supervised pretraining before attempting further architectural scaling.

Figure 6: Validation IoU (left) and F1 (right) across all four experiments. Red: enhanced baseline (`unet_aspp_model_improvement`), dark blue: baseline (`unet_aspp_dice_bce`), light blue: pretrained ResNet-34 (`siamese_resnet34_pretrained`), pink: SiameseUNet_V2 (`siamese_v2_attn_changefusion_deepsup`).

6.6 Test Set Evaluation

The best-performing model — the Enhanced SiameseUNet ASPP trained with Dice+Focal loss and data augmentation — was evaluated on the held-out test set of 867 samples. The decision threshold τ^* was selected by searching over $[0.05, 0.55]$ in increments of 0.05 to maximise F1-Score on the validation set; an optimal value of $\tau^* = 0.30$ was identified, lying substantially below the conventional value of 0.5.

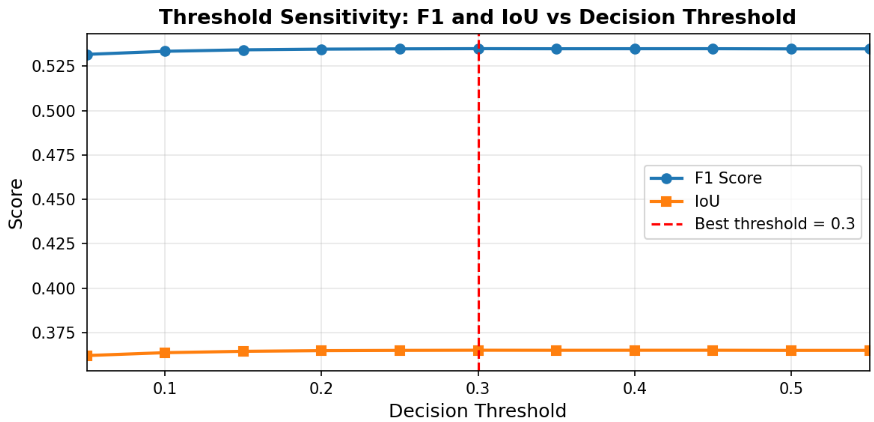


Figure 7: F1 and IoU as a function of decision threshold. Both metrics plateau across the range $[0.25, 0.50]$, indicating the model is robust to moderate threshold variation in this region.

6.6.1 Full Metric Breakdown

Table reports the complete pixel-level evaluation results on the held-out test set.

Metric	Value
Decision Threshold (τ^*)	0.30
Precision	0.4793
Recall	0.6049
F1-Score	**0.5348**
IoU (Jaccard)	**0.3650**
Specificity	0.9920
PR-AUC	0.5141
ROC-AUC	0.9776
True Negatives (TN)	55,690,792
False Positives (FP)	447,733
False Negatives (FN)	269,105
True Positives (TP)	412,082

Table 6: Test set evaluation results for the Enhanced SiameseUNet ASPP (threshold = 0.30).

6.6.2 Discussion of Test Set Results

The test F1 of 0.5348 and IoU of 0.3650 are consistent with the best validation metrics (F1 = 0.5189, IoU = 0.3526), confirming that no significant overfitting occurred and that the model generalises reasonably to the held-out distribution.

The ROC-AUC of 0.9776 demonstrates strong discriminative ability in probability space. However, the PR-AUC of 0.5141 — considerably lower than the ROC-AUC — reflects the fundamental difficulty of achieving high precision and recall simultaneously under severe class imbalance, where positive (forest loss) pixels constitute only 1.20% of all test pixels. This large gap between ROC-AUC and PR-AUC is a diagnostic signal: the model ranks positive pixels correctly in the ranked probability list, but cannot convert this ranking into precise binary predictions at any single threshold in figure 7.

The optimal threshold of $\tau^* = 0.30$ is well below the standard value of 0.5, indicating that the model assigns systematically lower confidence scores to forest loss predictions. This is a consequence of training on a dataset dominated by background patches: the posterior probabilities are calibrated to the prior, shifting the effective operating point downward.

The specificity of 0.9920 confirms that false positive predictions are rare in absolute terms ($\approx$0.80% of background pixels are incorrectly flagged), while the recall of 0.6049 indicates that approximately 60% of true forest loss pixels are successfully detected at the chosen threshold.

6.7 Per-Category Performance Analysis

To characterise model behaviour as a function of deforestation intensity, the test set was stratified into four categories based on the proportion of changed pixels per patch. Per-category pooled pixel-level metrics are

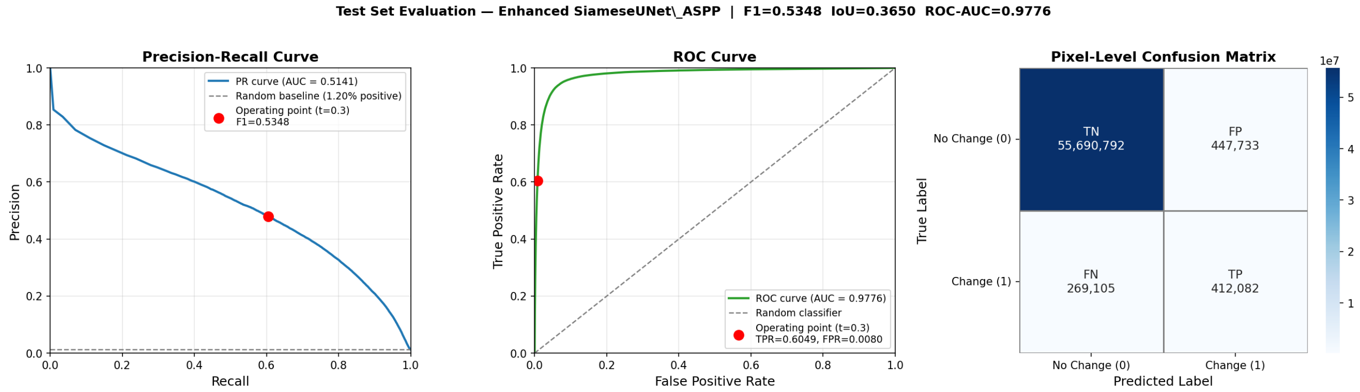


Figure 8: Test set evaluation plots. Left: Precision-Recall curve (AUC=0.5141) with operating point at threshold 0.30. Centre: ROC curve (AUC=0.9776). Right: Pixel-level confusion matrix. The operating point (red dot) reflects the selected threshold of 0.30.

presented in Table 7.

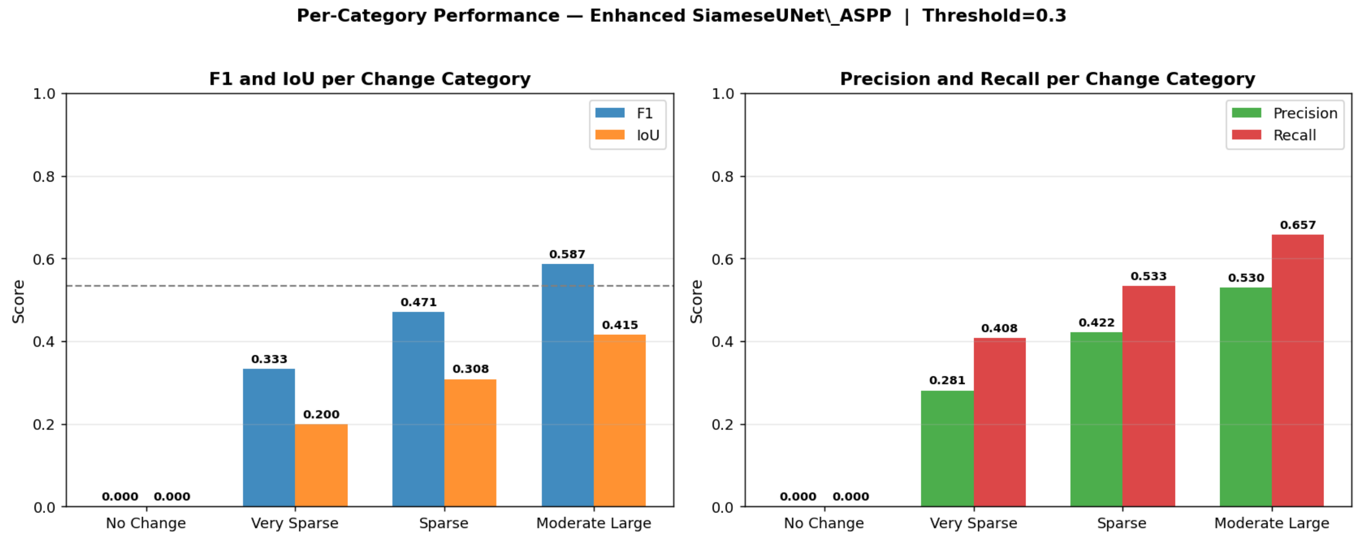


Figure 9: Per-category F1, IoU, Precision and Recall on the test set. Performance scales consistently with change intensity, with *moderate large* patches achieving the highest scores across all metrics.

Category	Samples	Avg GT px	Prec.	Recall	F1	IoU
No Change	97	0	0.000	0.000	0.000	0.000
Very Sparse	308	129	0.281	0.408	0.333	0.200
Sparse	301	690	0.422	0.534	0.471	0.308
Moderate-Large	161	2693	0.530	0.657	0.587	0.415
Overall	**867**	—	**0.479**	**0.605**	**0.535**	**0.365**

Table 7: Per-category pooled pixel-level metrics on the test set at threshold $= 0.30$. Avg GT px: average number of ground truth changed pixels per patch in that category.

6.7.1 Analysis

A clear and consistent monotonic relationship is observed between change intensity and model performance across all four metrics. The moderate-to-large category achieves the highest F1 of 0.587 and recall of 0.657, while very sparse patches yield an F1 of only 0.333 and recall of 0.408.

No Change Category. The no-change category produces zero F1 and zero recall by construction: these patches contain no positive pixels, so any prediction the model makes is by definition a false positive. The near-perfect specificity of 0.9997 for this category indicates that the model is appropriately conservative, rarely predicting forest loss in patches where no change occurred.

Very Sparse Category. With an average of only 129 changed pixels per patch ($< 0.5\%$ of the 65,536 pixels in a 256×256 patch), very sparse patches represent extremely small Jhum clearances. The low F1 of 0.333 in this category reflects the combination of insufficient spatial context — small patches do not provide enough surrounding pixels for the model to confidently infer the presence of a clearance event — and label noise at 10-metre resolution where 30-metre Hansen labels may not precisely delineate the boundaries of small disturbances.

Sparse Category. Sparse patches (0.5–2% changed, average 690 changed pixels) achieve an F1 of 0.471, representing a substantial improvement over the very sparse tier. With more changed pixels per patch, the model can exploit spatial context from adjacent changed pixels more effectively, improving both precision and recall.

Moderate-to-Large Category. Patches with over 2% changed pixels (average 2,693 changed pixels) represent the most detectable deforestation events — large contiguous clearances, road-associated clearings, and major Jhum openings. The F1 of 0.587 in this category confirms that the model is effective at detecting structurally prominent deforestation events, and the relatively high precision (0.530) indicates fewer false alarms compared to sparser categories.

6.8 Qualitative Results

Qualitative inspection of model predictions across all four change categories reveals consistent spatial patterns that align with the quantitative per-category analysis.

6.8.1 Moderate-to-Large Change Category

For patches with large contiguous deforestation events shown in figure 10, the model produces spatially coherent predictions that broadly capture the shape and extent of clearance areas. True positive detections (green in the error overlay) are clustered in the interior of large clearance events, where spectral differences between T1 and T2 are strong and unambiguous. False negative detections (blue) are concentrated at the *boundaries* of clearances — pixels at the edge of a deforested patch where the canopy transition is gradual and the spectral signature is mixed. False positives (red) cluster along terrain features such as ridgelines, river banks, and steep slopes, which exhibit surface characteristics (exposed rock, sparse canopy, moisture gradients) that mimic the spectral signature of deforestation.

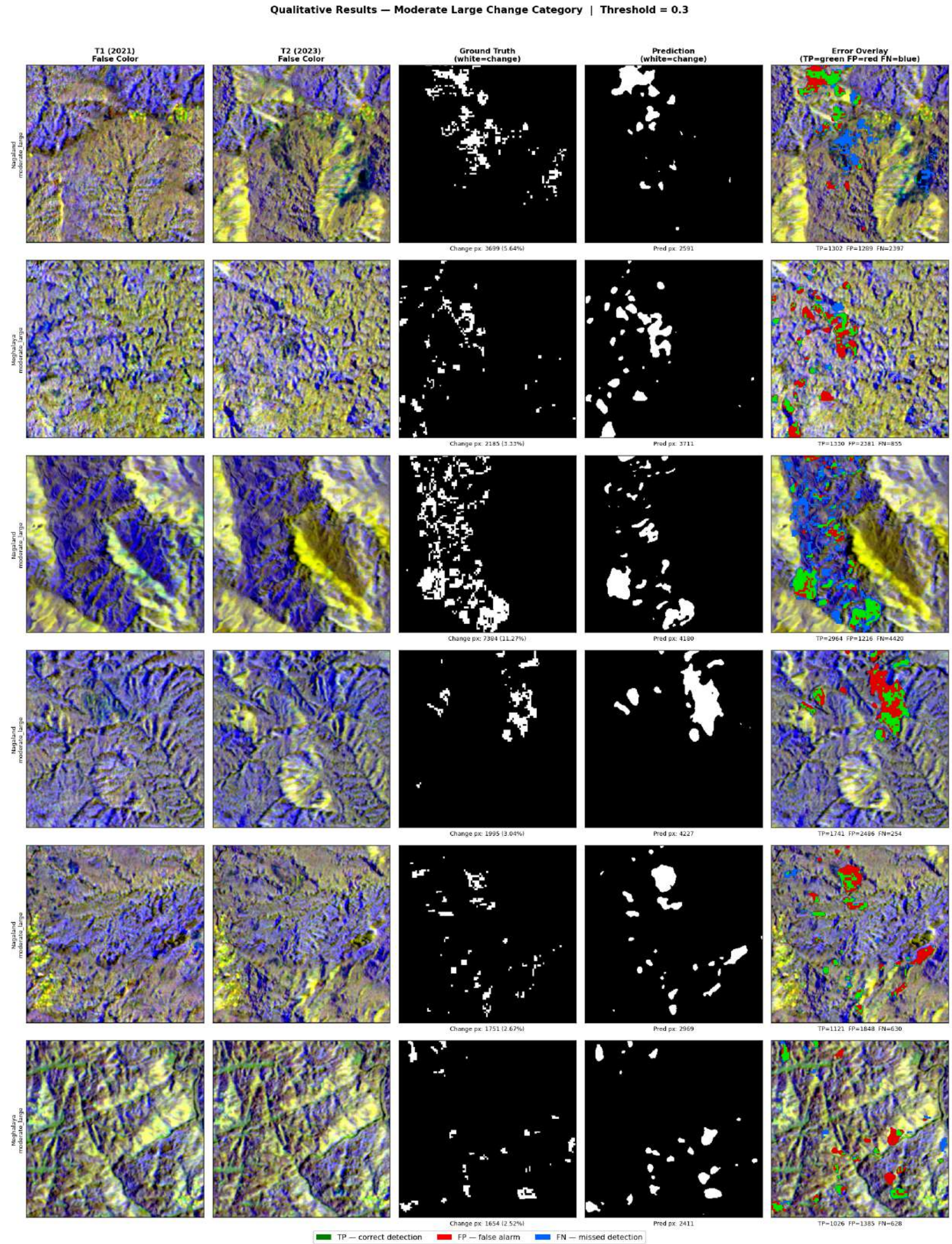


Figure 10: Qualitative results for moderate-large change. Large clusters detected (green), fragmented/boundary misses (blue), and false alarms (red) near transitions.

6.8.2 *Sparse Change Category*

In sparse patches, the model struggles to detect isolated, small-area Jhum clearances shown in figure 11. The majority of ground truth changed pixels appear as missed detections (false negatives, blue) in the error overlays. Successful detections are limited to clusters with sufficient spatial contiguity — isolated single-pixel or two-pixel clearances are systematically missed. This pattern is consistent with the U-Net decoder's receptive field behaviour: without sufficient context from surrounding changed pixels, the per-pixel classification is unreliable.

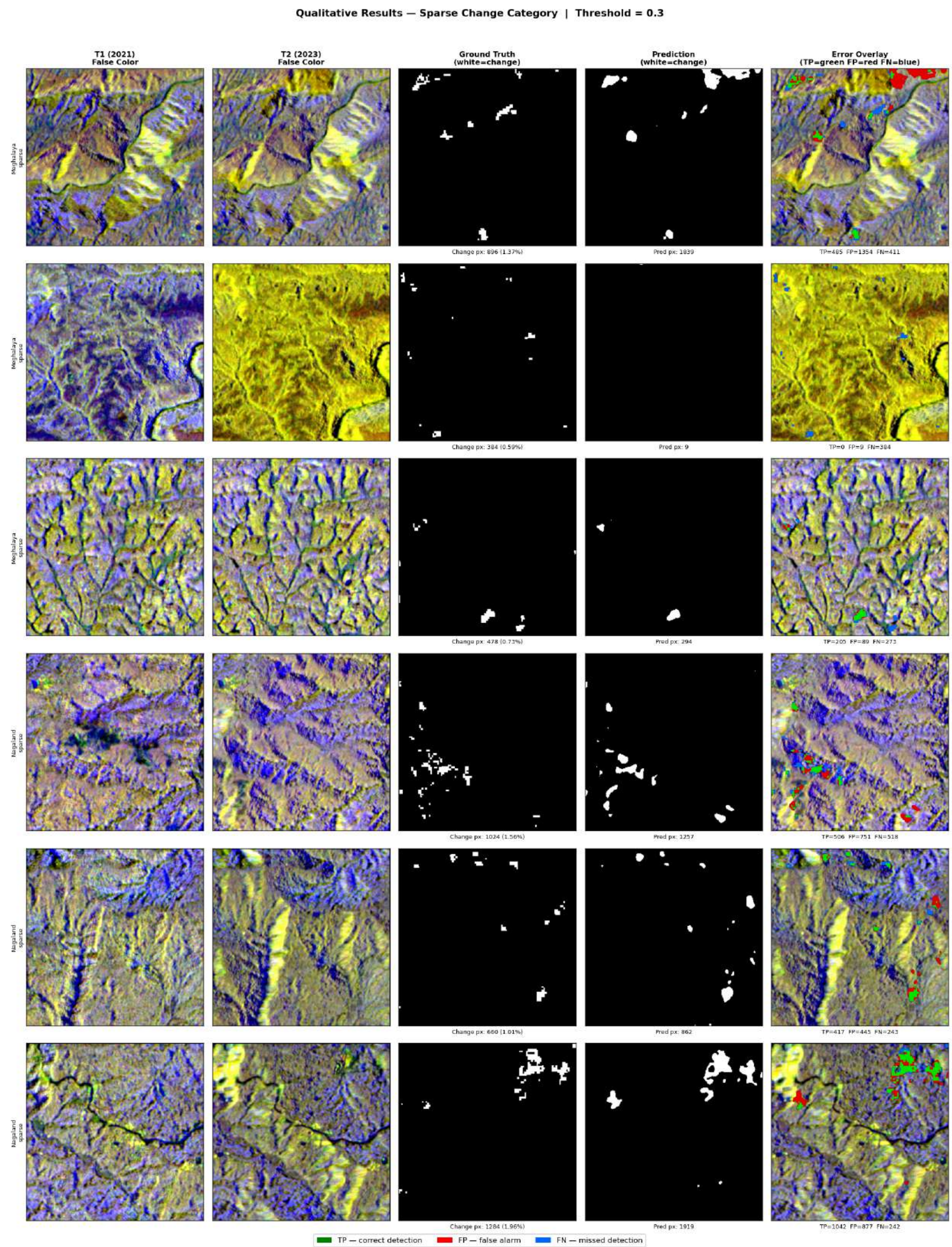


Figure 11: Qualitative results for sparse change. Small isolated patches are mostly missed (blue), with detections limited to more contiguous regions.

6.8.3 Very Sparse Change Category

Very sparse patches show minimal positive predictions. The model effectively treats these patches as “no change” in many cases, producing near-black prediction masks shown in figure 12. The few predictions that do appear tend to be false positives associated with terrain artefacts rather than true forest loss.

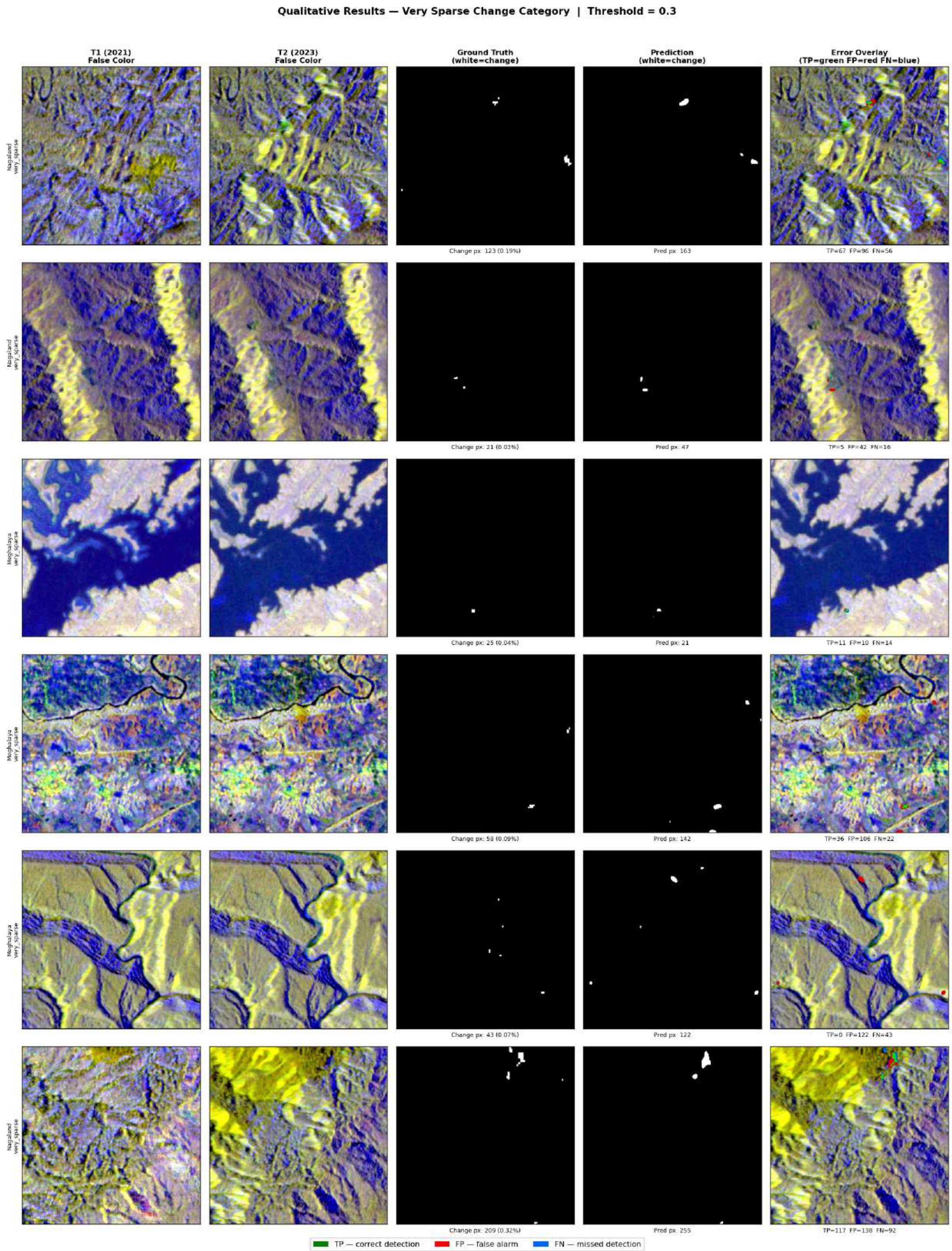


Figure 12: Qualitative results for very sparse change.

6.8.4 No Change Category

In no-change patches, the model’s output is consistently near-zero probability across the full patch, demonstrating robust suppression of false alarms shown in figure 13. This is consistent with the high specificity

(0.9997) reported in the per-category metrics.

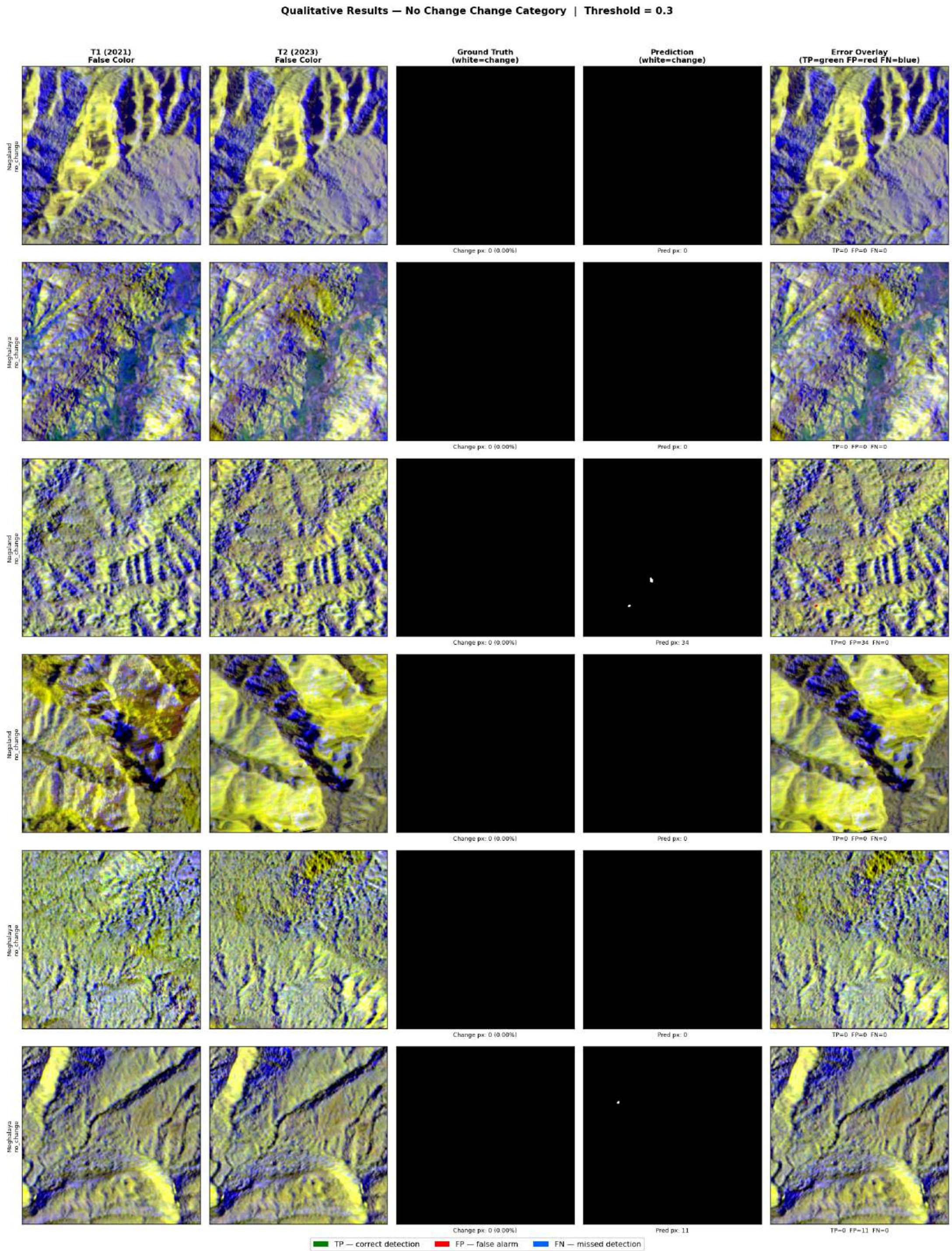


Figure 13: Qualitative results for sparse change. Small isolated patches are mostly missed (blue), with detections limited to more contiguous regions.

7 Error Analysis

Systematic analysis of the per-category metrics and qualitative visualisations reveals four recurring failure modes that together account for the majority of model errors.

7.0.1 Scale Sensitivity

The model's performance degrades substantially as deforestation patches shrink in spatial extent, with F1 declining from 0.587 (moderate-large) to 0.333 (very sparse). This is a direct consequence of the U-Net decoder's effective receptive field: Jhum clearances spanning fewer than approximately 50 pixels at 10-metre resolution (i.e., $<$5,000 m^2 or $<$0.5 ha) produce insufficient spatial context for the convolutional classifier to confidently assign the change label, particularly when the surrounding unchanged forest exhibits similar spectral characteristics due to mixed pixels at patch boundaries.

7.0.2 Systematic Under-Detection (High False Negative Rate)

Across all change categories, recall consistently exceeds precision, and the optimal threshold of $\tau^* = 0.30$ lies well below 0.5. This pattern indicates that the model assigns systematically low confidence scores to genuine forest loss pixels. The 269,105 false negatives versus 447,733 false positives in the overall test set confirm this asymmetric bias. Two mechanisms contribute to this: (i) *label noise* — boundary pixels in the Hansen-derived ground truth mask may correspond to only partially cleared pixels at 10-metre Sentinel-2 resolution, causing the model to learn ambiguous supervision signals at change boundaries; and (ii) *training imbalance* — despite Focal and Dice loss weighting, the overwhelming majority of patches are dominated by background, and the model's confidence calibration naturally skews toward low probabilities for the minority class.

7.0.3 False Alarms at Terrain Boundaries

Error overlays for moderate-to-large patches consistently show false positive clusters concentrated along topographic features: ridgelines, river banks, and steep terrain transitions. These regions exhibit spectral signatures that partially overlap with deforested areas — reduced canopy density along exposed ridgelines, bare rock at cliff faces, and moisture-driven spectral variation near rivers can collectively mimic the reduced NIR and elevated red reflectance characteristic of Sentinel-2 imagery over forest clearances. Without explicit digital elevation model (DEM) inputs, the model has no mechanism to distinguish topographic bare ground from anthropogenic clearances.

7.0.4 Spectral Similarity Between T1 and T2

Several patches display near-zero prediction despite unambiguous ground truth change. Visual inspection of false colour composites for these samples reveals minimal perceptible spectral difference between the 2021 (T1) and 2023 (T2) image pair. Two explanations are plausible: (i) *Jhum regrowth* — if a Jhum clearance was executed shortly after the T1 acquisition date and had substantially regrown by the T2 acquisition date in 2023, the T2 image will exhibit spectral characteristics nearly indistinguishable from the T1 forest, despite the Hansen label indicating loss; and (ii) *atmospheric normalisation* — seasonal median compositing during preprocessing may have reduced genuine spectral contrasts below the model's detection sensitivity threshold. This failure mode is a fundamental limitation of two-date binary change detection and motivates the use of multi-date time series in future work.

8 Discussion and Limitations

The overall test F1 of 0.5348 and IoU of 0.3650 represent a meaningful baseline for high-resolution, cloud-robust forest change detection in the NER of India, but remain below the project target of F1 > 0.80. The large disparity between ROC-AUC (0.9776) and PR-AUC (0.5141) is particularly informative: it demonstrates that the model discriminates well in probability space but cannot achieve simultaneous high precision and recall at the 1.20% positive pixel rate. This gap is a fundamental consequence of class imbalance rather than a modelling deficiency per se, and cannot be resolved by architectural changes alone without addressing the data distribution.

The key limitations identified through this evaluation are:

1. **Insufficient training data.** The 7,500-patch dataset is small relative to the capacity of modern deep learning models. Architectural complexity (pretrained encoders, attention mechanisms) requires substantially more data to yield consistent gains over simpler baselines.

2. **Label noise from resolution mismatch.** Hansen labels at 30-metre resolution introduce boundary uncertainty when applied to 10-metre imagery. Pixels at deforestation boundaries receive ambiguous supervision, inflating the false negative rate.

3. **Absence of cloud masking.** No explicit cloud masking was applied to the Sentinel-2 seasonal composites. Residual cloud and shadow contamination in the optical bands can introduce spurious spectral differences between T1 and T2 that the model misinterprets as deforestation events, contributing to the false positive rate at non-terrain locations.

4. **Two-date temporal approach.** The use of only two acquisition dates prevents the model from distinguishing permanent deforestation from seasonal or cyclical land-use changes such as Jhum regrowth. Multi-date time series incorporating 3–5 acquisition epochs would provide richer temporal context for disambiguating permanent versus transient canopy changes.

5. **Absence of topographic inputs.** False alarms at terrain boundaries could be substantially reduced by incorporating a digital elevation model (DEM) as an additional input channel, providing the model

with explicit topographic context.

9 Model Deployment and Real-Time Inferencing

9.1 Overview

To demonstrate the practical utility of the trained forest cover change detection system beyond offline evaluation, the best-performing model — the Enhanced SiameseUNet ASPP trained with Dice+Focal loss and data augmentation — was deployed as an interactive, publicly accessible web application. The deployment pipeline integrates Google Earth Engine (GEE) for live satellite data retrieval, a FastAPI backend for model inference, and a browser-based frontend for user interaction. The application is containerised using Docker and hosted on **Hugging Face Spaces**, providing a reproducible and scalable serving environment with zero infrastructure management overhead. The live application is accessible at:

https://huggingface.co/spaces/suranjan90/forest-cover-loss

9.2 System Architecture

The backend is implemented as a **FastAPI** application (`main.py`) that exposes a set of REST and Server-Sent Events (SSE) endpoints. The full request lifecycle proceeds as follows:

1. **Area selection.** The user draws a polygon on an interactive Leaflet map in the browser and selects a temporal window (T1 year and T2 year). A `POST /api/layers` request is dispatched to the backend with the polygon coordinates and year parameters.

2. **Satellite data retrieval.** The backend calls `inf.prepare_layers()`, which uses the authenticated Google Earth Engine Python API to construct Sentinel-1 SAR (VV, VH) and Sentinel-2 optical (B2, B3, B4, B8) seasonal composites for the specified polygon and years, and returns tile URLs for display in the browser alongside the map overlay.

3. **Tile prefetching.** A `POST /api/prefetch` request triggers asynchronous prefetching of the image tiles from GEE into a server-side session cache, identified by a unique session identifier (`sid`).

4. **Streaming inference.** The frontend opens a `GET /api/predict/stream` connection using the browser's `EventSource` API. The backend streams real-time progress events (tile download, normalisation, patch inference, result assembly) via Server-Sent Events (SSE), allowing the user to monitor prediction progress without polling. Model weights are downloaded on first request from the Hugging Face model repository (`suranjan90/forest-cover-loss`) using the `huggingface_hub` library and

cached locally to avoid repeated downloads.

5. **Result delivery.** Upon completion, the backend assembles the per-patch binary predictions into a full-resolution GeoTIFF-aligned prediction image, re-projects it to the map CRS, and returns it as a transparent PNG overlay via `GET /api/pred_img/{sid}`. A slippy-map tile endpoint (`/api/pred_tile/{sid}/{`
is also provided for seamless integration with the Leaflet tile layer system.

6. **Interactive re-thresholding.** The user can interactively adjust the decision threshold via a slider in the UI, triggering a lightweight `POST /api/rethresh` request that re-applies the new threshold to the cached probability map without re-running inference, enabling instant visual feedback.

9.3 Containerisation and Deployment

The application is packaged as a **Docker** container, ensuring full reproducibility of the Python environment, system dependencies, and CUDA runtime across development and production. The `Dockerfile` installs all Python dependencies (`fastapi`, `uvicorn`, `torch`, `earthengine-api`, `huggingface_hub`) and configures the `uvicorn` ASGI server to listen on port 7860, which is the standard port for Hugging Face Spaces. GEE authentication is handled via a service account credential injected as the `GEE_KEY_JSON` environment secret at deployment time, ensuring that no credentials are embedded in the container image.

The Docker image is hosted and served on **Hugging Face Spaces** under the `suranjan90/forest-cover-loss` namespace. Spaces provides automatic HTTPS termination, Nginx reverse-proxy buffering (disabled via the `X-Accel-Buffering: no` header for SSE compatibility), and persistent cold-start handling. A `GET /health` endpoint exposes runtime status including device availability (CPU or CUDA), enabling monitoring of the serving environment.

9.4 Summary

Table 8 summarises the key components of the deployment stack.

Component	Technology / Detail
Backend framework	FastAPI + Uvicorn (ASGI)
Satellite data	Google Earth Engine Python API
Model weights hosting	Hugging Face Hub (`suranjan90/forest-cover-loss`)
Inference framework	PyTorch (CPU / CUDA auto-detect)
Progress streaming	Server-Sent Events (SSE)
Frontend map	Leaflet.js with tile layer overlays
Containerisation	Docker
Hosting platform	Hugging Face Spaces
GEE authentication	Service Account (injected secret)
Live application URL	https://huggingface.co/spaces/suranjan90/forest-cover-loss

Table 8: Deployment stack summary.

10 Conclusion

This study presented a systematic investigation into AI-based forest cover change detection in the North Eastern Region of India, specifically targeting the Meghalaya and Nagaland states over a 2021–2023 temporal window. A multi-modal dataset of 7,500 bi-temporal patches was constructed by fusing Sentinel-1 SAR and Sentinel-2 optical imagery with Hansen Global Forest Change labels, and a baseline `SiameseUNet_ASPP` model was established as the quantitative reference point.

Four experiments were conducted in a controlled, step-by-step manner to isolate the contribution of each modification. The results demonstrate that training-level improvements — specifically the combination of Dice and Focal loss with data augmentation and dynamic threshold optimization — provided the most consistent gains, yielding the best observed F1 of 0.5189 and IoU of 0.3526. Architectural modifications including a pretrained ResNet-34 encoder and the attention-augmented `SiameseUNet_V2` did not surpass the baseline under the current dataset scale, highlighting that model complexity alone is insufficient when training data is limited.

Future work will focus on dataset expansion through additional geographic regions and temporal windows, exploration of semi-supervised or self-supervised pretraining on unlabelled satellite imagery, and integration of purpose-built change detection transformers such as BIT which are designed specifically for bi-temporal remote sensing tasks. A combination of the most effective strategies from this milestone will serve as the starting point for the final modelling phase.

References

[1] N. Myers, R. A. Mittermeier, C. G. Mittermeier, G. A. B. da Fonseca, and J. Kent, "Biodiversity hotspots for conservation priorities," *Nature*, vol. 403, no. 6772, pp. 853–858, 2000.

[2] M. D. Behera, P. Tripathi, P. Das, S. K. Srivastava, and P. S. Roy, "Assessment of shifting cultivation fallows in Northeastern India using Landsat imageries," *Tropical Ecology*, vol. 61, pp. 84–96, 2020.

[3] M. C. Hansen, P. V. Potapov, R. Moore, M. Hancher, S. A. Turubanova, A. Tyukavina, D. Thau, S. V. Stehman, S. J. Goetz, T. R. Loveland, A. Kommareddy, A. Egorov, L. Chini, C. O. Justice, and J. R. G. Townshend, "High-resolution global maps of 21st-century forest cover change," *Science*, vol. 342, no. 6160, pp. 850–853, 2013.

[4] B. Chen, B. Huang, and B. Xu, "Multi-source remotely sensed data fusion for improving land cover classification," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 124, pp. 27–39, 2022.

[5] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, “A tutorial on synthetic aperture radar,” *IEEE Geoscience and Remote Sensing Magazine*, vol. 1, no. 1, pp. 6–43, 2013.

[6] Q. Zhang, L. Ge, R. Zhang, G. I. Metternicht, Z. Du, J. Kuang, and M. Xu, “DSNUNet: An improved forest change detection network by combining Sentinel-1 and Sentinel-2 images,” *Remote Sensing*, vol. 14, no. 19, p. 5046, 2022.

[7] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, “DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 4, pp. 834–848, 2017.

[8] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.

[9] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[10] J. Xiang, Y. Xing, W. Wei, E. Yan, J. Jiang, and D. Mo, “Dynamic detection of forest change in Hunan province based on Sentinel-2 images and deep learning,” *Remote Sensing*, vol. 15, no. 3, p. 628, 2023.

[11] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional networks for biomedical image segmentation,” in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, ser. Lecture Notes in Computer Science, vol. 9351. Springer, 2015, pp. 234–241.

[12] R. C. Daudt, B. Le Saux, and A. Boulch, “Fully convolutional Siamese networks for change detection,” in *25th IEEE International Conference on Image Processing (ICIP)*. IEEE, 2018, pp. 4063–4067.

[13] H. Chen, C. Wu, B. Du, and L. Zhang, “A Siamese network based U-Net for change detection in high resolution remote sensing images,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[14] M. Papadomanolaki, M. Vakalopoulou, and K. Karantzalos, “A deep multitask learning framework coupling semantic segmentation and fully convolutional LSTM networks for urban change detection,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 9, pp. 7651–7668, 2021.

[15] European Space Agency, “Sentinel-1 mission overview,” https://en.wikipedia.org/wiki/Sentinel-1, 2023, accessed: 2026-04-02.

[16] Google Earth Engine, "Sentinel-1 SAR GRD: C-band synthetic aperture radar ground range detected," https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD, 2024, accessed: 2026-04-02.

[17] European Space Agency, "Sentinel-2 MSI level-2 processing," https://sentinel.esa.int/web/sentinel/user-guides/Sentinel-2-msi/processing-levels/level-2, 2023, accessed: 2026-04-02.

[18] Copernicus / ESA, "Sentinel-2 mission overview — SentiWiki," https://sentiwiki.copernicus.eu/web/s2-mission, 2023, accessed: 2026-04-02.

[19] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *International Conference on Machine Learning (ICML)*, 2015, pp. 448–456.

[20] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, and D. Rueckert, "Attention U-Net: Learning where to look for the pancreas," in *Medical Imaging with Deep Learning (MIDL)*, 2018.

[21] C.-Y. Lee, S. Xie, P. Gallagher, Z. Zhang, and Z. Tu, "Deeply-supervised nets," in *Artificial Intelligence and Statistics (AISTATS)*, 2015, pp. 562–570.

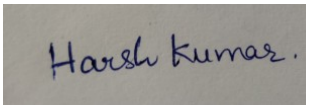
[22] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. Jorge Cardoso, "Generalised Dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, ser. Lecture Notes in Computer Science, vol. 10553. Springer, 2017, pp. 240–248.

[23] P. Micikevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh, and H. Wu, "Mixed precision training," in *International Conference on Learning Representations (ICLR)*, 2018.

[24] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *International Conference on Learning Representations (ICLR)*, 2015.

[25] Weights & Biases, "Weights & biases: MLOps platform," https://wandb.ai, 2024, accessed: 2026-04-02.

Signatures



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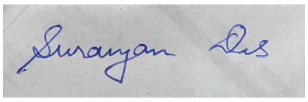

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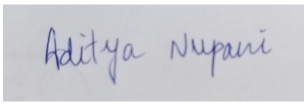

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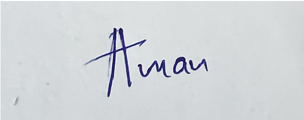

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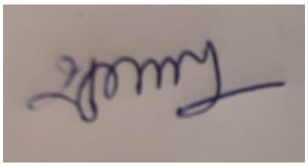

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